Faculty of Engineering

Module Title: Engineering Chemistry

Revision

A. Property: - any characteristic that can describe or identify matter.

- Properties can be classified as
  - **Intensive properties:** like temperature and melting point have values that do not depend on the amount of sample
  - **Extensive properties:** like length and volume have values that do depend on the amount of sample

B. Pressure: a force (F) exerted per unit area (A), where force is defined as mass (m) times acceleration (a)

$$P = F/A = (m \times a)/A$$

1 atm= 760 mmHg (torr)

$=1.01325*10^5$ Pa

$= 1.01325*10^2$ kPa

= 1.01325 bar

=76 cmHg

C. Volume: The amount of space occupied by an object

$1m^3=1000$ L$=1000$ dm^3

$1dm^3=1$ L

1L=1000 ml (cm^3)

D. Temperature:

$^0K = T\ ^0C + 273.15$

$^0F = 1.8\ T\ ^0C + 32$

E. Amount of substance:

Number of moles = mass/ molecular weight

Kmole=kg/M.Wt

mole= gr/M.Wt

mmloe= mgr/M.Wt

Kmole=10^3 mole, and mole =10^3 mmole

The fundamental and some derived units of measure

Physical Quantity	SI units	Fr units
mass	**Kg**	**gr**
Length	**m**	**cm**
Area	m^2	cm^2
Volume	m^3	cm^3
Time	**sec**	**sec**
Temperature	**K**	**K**
Density	Kg/m^3	gm/cm^3
Speed	m/s	cm/s
Acceleration	m/s^2	cm/s^2
Force	$N\ (kg.m)/s^2$	$gm.cm/s^2$
Pressure	$Pa\ (N/m^2)$	atm
Energy	$J\ (kg.m^2)/s^2$	cal "1 cal=4.18 J"
Amount of substance	**Kmol**	**Mol (gmol)**

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Sheet 1

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1. Typical atmospheric pressure on top of mountain Everest (29,035 ft) is 265 mmHg. Convert this value to pascals, atmospheres, and bars.
2. **Convert the following: -**
    - Pressure of 1150 torr to atm, Pa, cmHg, bar, mmHg
    - Carbon dioxide volume of 500 cm^3 to L, mL, $dm^3$and m^3
    - Temperature of 25 C to K and F
3. Calculate the number of moles of 50 gr of nitrogen gas, 640 mgr of oxygen

    [atomic weight:- N=14 O=16]

4. How many moles of methane gas (CH_4) are in a storage tank with a volume of 1.0×10^5 L at STP? And how many grams?

5. An Aerosol spray can with a volume of 350 mL contains 3.2 g of propane gas (C_3H_8) as propellant. What is the pressure in atmospheres of gas in the can at 20 °C?

6. A helium gas cylinder used to fill balloons has a volume of 43.8 L and a pressure of 1.51×10^4 kPa at 25.0 °C. How many moles of helium are in the tank?

7. What final temperature (°C) is required for the pressure inside an automobile tire to increase from 2.15 atm at 0 °C to 2.37 atm, assuming the volume remains constant?

8. **Define:**

- Boyle's law
- Charles's law
- Avogadro's law
- Mole fraction
- Ideal gas law

9. Choose the correct answer:

A. The Correct statement is:

- Gases at low pressure or high temperature behave as an ideal gas
- Gases at high pressure or low temperature behave as an ideal gas
- Gases at high density behave as an ideal gas
- None of the mentioned

B. What is the density of Carbon dioxide (CO_2) in kg/m^3 at 27 °C and 100 kPa?

- 1.76
- 2.76
- 3.76
- 4.76

C. Atmospheric pressure and 0°C conditions are known as:

- Room temperature and pressure
- Standard temperature and pressure
- Atmospheric temperature and pressure
- None of the mentioned

D. For an ideal gas, which pair of variables is inversely proportional to each other (if all other conditions remain constant)?

- P, T
- P, V
- V, T
- n, V

10. Uranium hexafluoride is a molecular solid used to purify the uranium isotope needed to fuel nuclear power plants. Assume that you have a 22.9 L vessel that contains 512.9 g of UF6 at 70.0 °C. What is the pressure in the vessel calculated using the ideal gas law and the van der Waals equation? (For UF6, a= 15.8 L^2.atm/mol^2, b= 0.1128 L/mol). (Molar mass of UF6 is 352.02 g/mol)